Question	Answer
Image Data why only 0= black and 255 White	For image data, generally 8 bits (256 different possible integers) are used to store the information. So 0 to 255 values correspond to colors from complete black to complete white. Therefore 0 is black and 255 is white.
Its 8 bit number system	You are absolutely correct.
8 bit.. 2 power 8 = 256 patterns	You are absolutely correct.
Does higher bits mean high quality image? If yes, how much can be the maximum bits?	Beyond 16-bit, the extra precision mostly helps computation for our data science problems for instance, not the visible quality.
Does higher bits mean high quality image? If yes, how much can be the maximum bits?	Yes, higher bits does in fact mean better quality image, more specifically because we have more number of different available values, so better separation in the colors or values. There is no limit to the maximum bits, but images generally use 8 bits to have good enough quality images along with low file size when saving.
If the input image has a hex color notation (six-digit code preceded by a pound sign, ex: Black: #000000), how is the conversion done to 8 bit number system? Kindly clarify.	So the first two places is Red, the third and fourth places is Green, and the last two places is Blue. Also, the symbols used are hexadecimal, which means each of the places can actually have 16 different values. Therefore, 00 is lowest and FF is highest, having a total of 256 different values for each channel. So for example, #FF0000 is completely red, with no green or blue.
what is the mechanism used to get a clear image out of a blurred image? How are the bits altered?	Deblurring works by estimating the blur pattern e.gmotion or defocus blur, it is called as "kernel" . USing which we can reverse that blur mathematically to redistribute smeared pixel values back to sharper edges. This is done through deep learning algorithms that recalculate each pixel's value based on neighboring pixels, sharpening transitions between them.
One generic question about this course - Its name is applied AI and ML. However, if we see literature around these terms, ML is a subset of AI. Why ML is mentioned separately from AI in this course?	today AI is used synonymously with claude, chatgpt etc. We specifically called out ML because our course is (As you see) designed to get into nuances within ML algorithms
Are there any attempts to apply lossless compression techniques, such as Huffman encoding, to image or video data for further bit-level optimization? How feasible is this in the machine learning domain, and what are the main challenges or computational complexities involved?	Yes — Huffman coding is already a core part of lossless/lossy pipelines. e.g : JPEG uses it after DCT quantization, PNG uses DEFLATE. In ML, learned entropy coding is an active research area, like NueRAL Netwoks learn compact latent representations, then a pre- TRained probability model feeds an entropy coder for optimal bit allocation
Are the libraries available for converting blurred images into deblurred images?	yes , cv2, ski.image, scipy etc
what package is there for higher precision image and image deblurring	scikit-image.restoration or OpenCV
How does the software keep track and differentiate pigs, all pigs almost look similar	like human faces, even animals, birds have difference in features that may or may not be visible to human eye. But there are nuances in the images that can be calculated when we look at pixel level. So the software can differentiate using the calculations
When a deblurring model identifies a blurry object with high confidence, does it recover the true original details or add false details to the image?	It is possible that false details are added, and in fact this happens a lot.

How does a system distinguish between a cycle and a bike from a blurry image when they have similar or different shapes?	Depends on how the system is made or trained. If it is trained to look for shapes, it does so, given that it is trained properly.
General question - Is creating small models included in this course?	yes of course, I saw regression, classification (basic models as well)
in color video analysis example of pigs, how do we differentiate between distressed and non-distressed ones? is there any heat characteristics or behavioral aspects which is taken into consideration?	Yes, these features, along with many other features are in fact used in combination.
Can we not use QR code instead as Qr code scanning is easier than processing the printed text images ?	QR codes need the content encoded beforehand , they can't read text that's just naturally printed, like a book page or sign, they work great for pre-palled activities like tickets, product IDs, tracking labels — but not for reading arbitrary existing text.
We can perform image classification with training data both with ANN and CNN	Both Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) can theoretically be used for image classification, but CNNs are the standard approach. Standard ANNs flatten images into a 1D array, which destroys 2D spatial relationships and requires a massive number of parameters for every pixel, leading to severe overfitting. CNNs solve this by using convolutional layers with shared weights to capture spatial hierarchies and visual patterns efficiently.
Is there any real life example that combines AI (Claude, Chatgpt, Gemini, etc.) and ML both to give the output?	Yes, in fact in real-life many AI models, including ML are combined.
But in handwritten pincode, it may not be horizontal line at bottom ?	you are right so that is why training should be done with handwritten data as well
can we not compare pixel positions and sequence	we are taking BOTH pixel positions and sequence as I/P vars
The email invite is for "Data analysis, pre-processing and visualization, ML workflow, cross-validation, error types and performance metrics". Why is the video about Image Recognition?	this is an example of pre-processing image data, we have other models coming up later in the day
What if we train it 100 images learn about and then check	100 images is usually less for training, the model may or may not work accurately with such low training data, it is always best practice to have many more images for training like 10K
When we say example based trainining, we give n number of data and say it is number 2 for example? How is the training code written?	yes the training data would have the answers also, many handwritted and typed entries of 2 and the corresponding dependent variable which says it actually is 2. So the model learns how 2 looks like across many entries so that when it encounters 2 in the live data it can detect correctly
If the model learns from all the images of the number 1 and is presented a different version of 1 with an underline in it, how does it rightly inference it even when none of the training images matched it ?	In general, the model will actually have difficulty detecting 1 with an underline correctly if it is not trained on 1 with an underline.
Is this how OCR has developed? Or that's a different concept?	Yes, OCR historically used image processing techniques, but now many AI-related techniques are used in combination with the traditional techniques.

does OCR packages use - this rule based features or machine learning models? - Digitization is important factor here	earlier OCR packages were using rule based features but now the Modern OCR like Tesseract 4+, EasyOCR, PaddleOCR etc primarily uses ML models CNNs + RNNs/Transformers trained on huge datasets of fonts/handwriting and this lets them handle varied fonts, noise, and blur far better than rigid rules.
@renuka S_TA: In the black and white image with scale 0 to 255 given example. Some white squares are also off 255. And full dark squares are away from 0. The image and the imposed numbers are not matching.	that was shown as an example , just look at the highlighted numbers for your understnading, we will check if the numbers and the colors are not aligned in the backend
Can we not use the rule of 1,2 refers north, 5,6 refers south to just classify the pin codes?	such rules may not always be true
Is there any thumb rules to understand that which problem can be solved by AI, which one could be solve without AI? Or can you let me know how we can build our thinking when a problem can be solve using AI?	This is a very nice question, and the entire cohort will be slowly developing the learners' intuition on where to use AI, and if so, which AI.
so the labeled dataset also has the answers?	yes training data would have the answers
How can we decide the label based on image, will this be a manual activity?	Yes, in general this is done manually. This is why creating datasets is also takes a lot of effort.
How Model indicate the pincode on postcard,there might be many such numbers	Each of the numbers can be detected one-by-one, and then each one can be classified, and all digits then can be combined to get the final code.
In this case, should the training be on every single type of RGB combination or will the background color be a trained to be ignored in real-world?	there are so many pre-trained models out there which we reuse using python packages and libraries, they are well trained on all of the RGB combo, background colors etc. e.g VGG, RESNET. WE can used them off the box
If we have to define AI today, what would be its definition? Is AI limited only to Chatbots (Claude, Gemini, ChatGPT, etc.)?	A lot of things, starting from an air conditioner detecting the temperature and turning itself on or off, to video generation, comes under AI, according to theory. But in day to day life, when people talk about AI, they mean the generative AI like chatbots and image/video generators.
What comes in out of distribution? Like we will not test with the diversed data? Blue background and black colour one	the pre-trained models in Python like Resnet, VGG and others are well tested on many diversed datasets so you do not have to worry about the accuracy
Renuka, This practice was not comunicated by your counceller while taking the admission for the course. You should have offered this course to only who has prior data science knowledge	let me take this up with the team and be assured your concerns would be taken into consideration. Thanks for brining it up
why are the testing data is in 1D , did not get it	think of it as the data getting flattened to 1D as the models like NNs understand 1D better. It is an internal process and is already part of the algorithms you do not have to do it as a step
is it possible that the system reading the input read every thing as white/ grey background and Black letters	Yes, it is completely possible.

Is Invoice extraction examples and extracting fields from invoices kind of Model will be build in this course?	thsi specific example may not be catered to in this course, however worry not, there are many python libraries availabe to do this easily. E.g : LayoutLM / LayoutLMv3 (Hugging Face transformers) — pretrained specifically for document understanding, extracts key-value fields (invoice number, date, total) using both text + layout position. Donut (Document Understanding Transformer, Hugging Face) — OCR-free, directly extracts structured fields from invoice images. PaddleOCR + PP-Structure — has dedicated invoice/table extraction pipelines.
raj , tell me about , why we are not use more then 70% of total data set	There is no hard limit to the percentage of data used for training. However, the dataset is generally divided into subsets to confirm that the trained model is actually usable in real life. For example, the training subset can be 70%, on which the model is trained, there can be 15% validation subset, which is used to confirm that the model can perform well on the data it is not trained on. Finally, the rest 15% can be used as testing subset, that is used to confirm that the model after the complete training period, is finally well suited to real-life examples. 70%-15%-15% is not necessary. For example, the splitting can be done as 80%-10%-10% as well. These concepts will be discussed in more detail in the next set of lectures.
How will it recognise a writing in different colours ?	using RGB algortithm prof walked us threough early on
How are images calculated?	using pixels
reject that bad sample and go for second round or manual verification	yes
How to decide how many types of Variance is needed?	you decide how many types - e.g rotation, flip, brightness, blur, crop, noise etc. based on what real-world variations your deployed model will actually face e.g. cars photographed at different angles/lighting need rotation+brightness variance, not necessarily color-shift.
Hello Renuka & Raj -- can you please let us know how the basic of ML can be learnt first....I feel Mahesh Sir is going too fast ...can we get some reading material	sure we will take up your concern with the team and have it heard. Thanks for bringing it up
I am totally disappointed.	I am sorry you are. But do give it some more time before giving up, prof Mahesh is doing such a good job in explaining from basics.
Can this be used to auto fill Capcha Codes? or identify Capcha values. I'm not Robot	that works differently. Capcha captures the movement of your mouse before the entry, Capcha doesn't get auto filled
training data will still be required as same query would come ? isn't it ?	training data is required only for training
what is the resoultion of the ink smudge or the bad samples ?	Could you kindly elaborate your question?
Mahesh Sir is talking al lot of techniques and examples...if this is summarized ...it will help to revise as well.	we will see what we can do, we will share materials

what is indistribution and out of distribution?	In-distribution: Data that comes from the same statistical distribution as the training dataset. The model has learned these patterns and can predict them reliably. Out-of-distribution: Data that significantly differs from the training dataset. The model has not encountered these types of examples during training, which typically leads to unpredictable or inaccurate results.
Can we use another approach - like use the pattern for 8 and scan through the image to see if the pattern is available with certain invariance in the entire image as a segment	Yes, and in fact this is used in traditional image processing techniques.
Can we make an accurate model to predict the stocks price (based on different factors) with the use of AI and or ML? Is this practically possible?	Predicting directional stock movement with high accuracy is not practically possible. Financial markets are incredibly noisy and heavily influenced by random, unpredictable events (Efficient Market Hypothesis). While ML models can analyze trends, sentiment, or historical patterns, they cannot reliably predict exact future prices or achieve accuracies like 80-90% due to the inherent randomness of the market.
RGB?	Red ,green , blue dimensions of colors. (255,0,0) being red for eg
what is loss function?	Loss function is a mechanism to let the model know how it is performing. For example, say images of digits are classified as 0 or 1 and so on. So in a simple sense, there can be a counter to calculate when the model detects wrongly (0 as 1 for example). Afterwards, this loss function (which increases as the error of the model increases) can be used to train the model so that it makes less of such mistakes. Please note that the type of loss that I gave as example is easy to understand, but practically other loss functions are used as they need to be able to inform the model properly. The corresponding mechanisms will be covered in later lectures in this course.
why in standard grayscale image we have 8 bits representation. if we use 4 bits can it be represented? how will it change in RGB, is it only impacting the channels or even bits	A 4-bit representation is technically possible and provides 16 discrete intensity levels per channel, compared to the 256 levels in an 8-bit image. Reducing an image to 4 bits therefore reduces the filesize needed, by compromising with the image quality (there are lesser number of different grayscale values possible).
How we are going to work on data which are protected or restricted? Is there any way to deidentified data for the ML and feed them for analysing ?	If the restricted data is available to the user (for example a working professional inside a company having the data), the identification information can simply be removed so that the model does not have that information (this comes in the preprocessing stage). If outside the company, the user does not actually have the data, and so missing data needs to be handled. Working around the missing data problem will be covered later in this course in the next few lectures.
sir we have data but how that gets labelled	In general, the labelling has to be done manually.

How should we decide volume of Training data to be used?	depends on how accurate you want the model to be to the varied new data. You can use pre-trained models actually if you do not have time or enough data to train
generally historical data are not labelled i have data let suppose 100K customers send post from one location to another but its not labelled	In general, the data still has to be manually labelled, making the dataset preparation also a task needing a lot of effort. Alternatively, AI models can be used to understand the context and label the data, but care should be taken so that the model themselves do not create too many wrong labels.
How to access your labelled data to make sure it's correct? Are there any techniques to analyze the data?	It depends very much on the data itself, but to check the correctness of labels should be manual. Alternatively separate AI models can be used to automate the analyzing task, but care should be taken so that the analyzer itself is not wrong.
Why is it 784 input Neurons?	If the input image is 28 pixels by 28 pixels, the total number of pixels is 28x28 = 784. Therefore 784 input neurons are needed for the 784 input values.
What is the difference between trainiung data and testing data	There are three subsets of data used in general. Training subset, validation subset and testing subset. Training subset is used during training to actually train the model. The validation subset is used to evaluate the model (during training) to ensure that the model can also work on unseen data. Finally, the testing data is used after training to confirm that the model can be deployed for unseen real world data.
If I get a dataset from Kaggle(or other sites) and models from HuggingFace(or other sites). both may not be compatible, correct? so there should be some data massaging or an interface layer that adopts the dataset to suit the model used. Is my understanding correct?	Yes, you are absolutely right. This stage is called preprocessing.
Is there any regulatory body, international organization, or standards-development organization that defines standards or frameworks for AI research, development, testing, evaluation, and deployment, similar to how 3GPP defines standards for mobile telecommunications and IEEE develops standards for various technologies	ISO/IEC JTC 1/SC 42 — the main international standards body for AI, produced standards like ISO/IEC 23894:2023 for AI risk management guidance, plus ISO/IEC 42001 (first certifiable AI management system standard, published Dec 2023). arxiv NIST AI Risk Management Framework (US) — organizes AI risk work into four functions — Govern, Map, Measure, and Manage — voluntary in the US but widely used as a baseline. Kong Inc. IEEE — has AI-specific standards too (e.g., IEEE 7000 series on ethical/algorithmic design). OECD AI Principles and UNESCO's Recommendation on Ethics of AI — international soft-law guidelines, voluntary but influential across G20 countries. arxiv EU AI Act — the closest to binding regulation (not just a "standard"), a risk-tiered law now entering enforcement phases through 2025–2026

what is indistribution and out of distribution?	In-distribution: Data that comes from the same statistical distribution as the training dataset. The model has learned these patterns and can predict them reliably. Out-of-distribution: Data that significantly differs from the training dataset. The model has not encountered these types of examples during training, which typically leads to unpredictable or inaccurate results.
what is ANN architecture	ANN stands for artificial neural network. It will be discussed in more details in the upcoming lectures.
So Loss Function in simple terms is where not go in future, in a way	it helps us identify the lowest point beyond which we should not go as its not required, once you hit rock bottom , why will you keep going ahead
What is softmax?	it IS a function used in the final layer of classification models that converts raw output scores logits into probabilities that sum to 1, so each class gets a probability, and the highest one is the predicted class.
what are logits?	raw output of the model which needs to be converted to 0 to 1 so that we understand that as probability
what is logits ?	think of it as a number without any range, we need to restrict the range to be between 0 and 1 as that is how we get the probabilitoies
Are there python libraries to derive these loss calculations? The mathematics formula is going over the head.	of course. scikit learn has them all don't worry
why we selected this specific loss function?	Depending on the task, the loss function changes. This will be detailed more in the upcoming lectures.
how to indetify which loss function to use ? Would that be covered ?	depends on the model type Binary classification → Binary Cross-Entropy (Log Loss) Multi-class classification → Categorical Cross-Entropy (or Sparse Categorical Cross-Entropy if labels aren't one-hot encoded) Multi-label classification (multiple correct labels per sample) → Binary Cross-Entropy applied per label
can we pause and understand the loss functiona equation, it was too fast	we will not be able to cover the entire material if we do so, thanks for your understanding .
Will the loss function change for every NN?	no, loss function depends on the task (classification, regression, etc.), not the network architecture itself. A CNN, RNN, or Transformer doing the same type of task (say, binary classification) will typically use the same loss (Binary Cross-Entropy).
How to find out which libariers to use for this mathematical calculations ?	numpy, matplotlib, and scikit learn has all the required functions
do we need to maintain strict distinction between loss function vs objective function or either of them is fine?	They're often used loosely/interchangeably in practice

Can we say regression is Analog signal vs classification is digital?	While this is true, the two (regression and classification) cannot be taken as analog/digital versions of each other. The reason is that while regression provides output values in continuous range, classification is not necessary done for outputs in a range (for example, classification into cats, dogs and birds cannot be assigned any discrete numerical value).
@renuka please do sugguest the book or refrence material for the session if thats possible	https://www.geeksforgeeks.org/deep-learning/neural-networks-a-beginners-guide/
what is cross entropy	measures how far off a model's predicted probabilities are from the actual true labels . The bigger the gap between predicted confidence and reality, the higher the loss.
Thanks @renuka Or can we have some books to start from basics to advance thats possible of link to reference material	yes , you can refer to this : https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html
why grayscale image is always 2D?	A grayscale image has some width and some height. Therefore it is 2D. In case of color image, there are multiple of these 2D images , therefore it has an additional dimension for the 3 2D data, namely the channel dimention for R, G and B
how can we label a dataset for a computer, and how can we train the computer to understand the data so that it can produce results that are as close as possible to the actual image?	Labelling is generally done manually. The models on a computer can be trained to understand the data by calculating the losses and letting the model train by learning from mistakes. More details to be given in the upcoming lectures.
Why is Grayscale Image data 2 dimensional? Assuming a 8 bit It has data between 0 and 255, one value per bit. Is that not a 1 dimensional data per bit?	2D is the spatial representation not bit denotion. Each pixel's 8-bit value (0–255) is just one scalar number — a single point of data — regardless of how many bits it takes to store that number. The "dimension" comes from how these scalars are arranged: a grayscale image is a 2D grid/matrix of such values (row, column) → hence 2D data, with one channel (depth of 1) at each position, using 8 bits to represent each pixel's intensity.
Hey just to get an intuition to plan the learning path. All the Math used here, do we need to know how to solve it on pen and paper or do we need to know how to use it in coding and the program does it for us? If I clear my concept of ML workflow (what to do when) do I need to know this computational part of the math or that will be handled by python	nope it will be handled by python. But good to know what is going in the backend
Also, was knowledge of Data science a prerequisite for this course?	
why do we need logit , softmax , relu function in neural network?	think of it as encoding the data into algo understandable format which has to be later decoded. Logit and relu for encoding, softmax for decoding
how does AI identifies different parameters like image has color and shape?	Different AI models (ANN, CNN) use different parameters/techniques to learn color/shape and so on. ANN, CNN and other models will be covered in details in the upcoming lectures.
Can you explain basic structure of ML, I am totally new to this area	think of it this way, you are using the known information to predict the unknown. Known data is fed into the model to train it so that it knows how the new data would react. Like our brain eg we learnt to walk , talk using this methodology unbeknownst to ourselves

Does RGB image helps in understanding variance more accurately - specifically color shift - or the angle shift in the camera click	RGB helps significantly with color-shift variance like ighting changes, white balance, hues etc since it captures actual color info across 3 channels. For angle/camera shift, RGB doesn't add much as that's a spatial/geometric problem, handled by data augmentation (rotation, perspective transforms) regardless of color or grayscale.
How can we decide the formula/ algorithm to fit into from raw data?	Depending on the use case , if you want to classify into a category it will be a classification model like decision tree or even ANN with softmax, if it is a number prediction it will be regression
Can we get list of question for interview related to these topocs	interviews will focus on applications rather than concepts. So depending on your area, please prepare the applications related Qs accordingly
What concepts comes under deep learning?	Deep learning includes concepts like computer vision, universal function approximations, generative chatbots, image generation, video generation.
Can we get a thumb rule between no of images and accuracy and class variance?	It depends on the data itself, so it is impossible to get a rule of thumb.
How can system distinguish between monkey man and gorilla where picture is not clean	the models are so sophisticated that they can unblur to an extent and can learn a ton from the pixels and the relationship between every pixel
Can ML identify if its partial image?	yes
Any block diagram which can explain how the ML works?	https://www.labellerr.com/blog/machine-learning-steps-a-complete-guide-to-the-ml-process/
How 1 and I are differentiated while training the system?	This has to be specifically included into the training data. For example, all 1 have a horizontal line at the top, whereas all I have the top line missing. However, this means that 1 written without a top line will be wrongly classified. In that case, the context of the sentence, if any, has to be included, using sentiment analysis. If these mechanisms are not used, the model is not able to differentiate 1 and I.
How can we use AI in database operations where it has already many unbuild tool. How it is defferent	AI doesn't replace the built-in tools like indexing or query planners, it just adds a smarter layer on top. Traditional tools follow fixed rules someone coded in, while AI learns from actual usage patterns and adapts over time, like predicting the best query plan based on real load instead of static logic. So things like auto-scaling, anomaly detection, and natural language to SQL are basically AI making the database self-improving rather than just automated. The real difference comes down to fixed rules versus learning and adjusting from real data.
How number 1 and I (letter I) are differentiated while training the system?	This has to be specifically included into the training data. For example, all 1 have a horizontal line at the top, whereas all I have the top line missing. However, this means that 1 written without a top line will be wrongly classified. In that case, the context of the sentence, if any, has to be included, using sentiment analysis. If these mechanisms are not used, the model is not able to differentiate 1 and I.

Explain use of AI in database operation	AI is making databases smarter and more hands-off. It predicts better query plans, auto-tunes performance, and catches unusual access patterns that could signal security issues. Tools like Oracle Autonomous Database and Amazon Aurora now auto-scale and self-heal without a DBA stepping in every time. On top of that, natural language to SQL tools let people just ask questions in plain English instead of writing queries, and AI also helps clean messy data and predict storage or failure issues before they become real problems.
Are we using always Vector & Linear Algebra concept underline all algorithm ?	Yes, we heavily use vectors and linear algebra, along with probability and statistics for AI based techniques. However, not all algorithms actually need vector and linear algebra.
If the model learns everything from ID data, wouldn't it be right to say that the model has memorized and will not perform well during inference ?	Yes, and this is called overfitting. To minimize this problem, the training is done with a subset of data, and the other subsets are used for evaluating the data to see how well it performs on unseen data.
While accessing some secure sites, the system asks human to identify images and things, to prove they are not robots? Why does ML fail here?	Machine learning models struggle with image CAPTCHAs because the images are intentionally modified with adversarial noise, severe distortions, or overlapping elements designed specifically to confuse visual models like Convolutional Neural Networks. Furthermore, CAPTCHAs frequently require complex contextual reasoning (e.g., identifying a crosswalk partially hidden behind a tree) that current models struggle to generalize without highly specific training.
Can we have a simple note what come sin ML and what come in DL	ML is the broad field where models learn from data, things like decision trees, random forests, SVM, and clustering, and it usually needs someone to manually pick out useful features first. DL is basically a subset of ML that uses multi-layer neural networks, like CNNs and Transformers, and it learns the features on its own straight from raw data. DL tends to need way more data and compute power, but it really shines on complex stuff like images, speech, and language. Simple way to think about it, ML is the umbrella, DL is just the neural network branch under it.
So when Google photos used to be free and unlimited, it collected all the samples and made it in such a way that all of the possible combinations of photos made it IN-DISTRIBUTION and a very little in Out Of Distribution. So, thier AI algo's work so well. But, how come the new AI's work almost as well as Google's?	they are built on top of pre-trained models like lama, gpt, etc

How do we define and classify the fitment of a loss function for a particular problem?	This comes from intuition and understanding of the mechanism of those particular loss functions. The loss functions will be covered in more details in upcoming lectures. However, if totally unsure, just train the model on multiple loss functions and finally select the one which performs the best on unseen data.
Is this Hyperparamemeter tuning , which reduces loss function value ?	Not quite, that's actually gradient descent doing the heavy lifting during training, adjusting the model's internal weights automatically to reduce loss. Hyperparameter tuning is a different, higher-level step. It's about choosing settings that control how training happens, things like learning rate, batch size, number of layers, or number of epochs, usually set before training starts and tweaked through experimentation. Good hyperparameters help gradient descent converge faster and reach a lower loss, but they don't directly reduce the loss themselves, they just make the actual learning process more effective.
Is it the concept of Gradient Descent , we are applying in loss function	yes Gradient descent is the optimization technique used to minimize the loss function, it calculates the gradient of the loss with respect to each model parameter and nudges the weights in the direction that reduces the loss. It keeps doing this step by step, adjusting weights a little at a time, until the loss gets as low as possible or stops improving. So loss function tells you how wrong the model is, and gradient descent is literally the mechanism that uses that error to correct the weights.
Again, in simple terms, what the machine is trying to do is learn and register feedback. After millions of trial and error and learning, there comes a point when it simply knows what all is right and what all is wrong and it comes close to 1.... Right?	yes The model keeps making guesses, checking how wrong it was using the loss function, and adjusting itself a tiny bit each time through gradient descent. Do that millions of times across tons of examples, and it slowly gets better at recognizing patterns, so its confidence for the right answer creeps closer to 1 while wrong answers drop closer to 0. One small nuance though, it's not really "knowing" in a human sense, it's just getting really good at statistical pattern matching from all that repeated feedback. But yeah, your instinct about it converging toward confident correct answers over time is right.
How do you define a parameter ?	already answered on the hyper parameter
Loss = F(W), what is W in this? Win??	W is "weights". In simple terms, the weights define how much of the input affect the output.

what is difference between AI,ML,Deeplearning and neural networks how it is useful in real world with examples?	AI is the broad goal of making machines act smart, think of things like Google Maps route optimization or Bing search suggestions. ML is a subset of AI where machines learn from data instead of being explicitly coded, like Netflix recommending shows based on your watch history. Deep Learning is a subset of ML using multi-layer neural networks, powering things like face unlock on your phone or ChatGPT-style text generation. Neural networks are literally the building blocks DL is made of, layers of interconnected nodes loosely inspired by the brain, used everywhere from voice assistants like Siri to self-driving car vision systems.
somehwere in the descent, we might run into local minima , how to ensure to reach to global minima?	using optimal learning steps and learning rate
if we only evenr find local minima, is it so that we can never train a best optimized model?	Yes, and this in fact happens a lot. In general for practical applications, there is no way to confirm that the model has converged to the global minima. Additionally, the global minima depends on the data, so the real-world data the model sees after the training can have a different global minima (slightly different in general if the data is similar).
Can there be multiple paths a gradient descent can take and hence have a different loss function?	yes
In this analogy, should we decide the direction of descent based only on the immediate local gradient, or should we consider the trend over a longer distance to determine whether we are genuinely moving toward a minimum?	Great question, and there are multiple optimizers which use both the approaches. In general however, a trend over a longer distance works better, and is used mostly today.
Can there be multiple minima for a loss function?	yes multiple local minimas and a global minima
how do we define weights	Weights are the numbers a neural network multiplies with each input to decide how much influence that input has on the final output, basically how strongly one neuron's signal affects the next neuron. During training, gradient descent keeps adjusting these weights based on the loss, higher weight means the network thinks that input matters more, lower or negative weight means it matters less or pushes the opposite way. Think of it like a dimmer switch on each connection between neurons, the network learns the right dimmer settings through millions of examples so it can turn up the important signals and turn down the noise.

In the image we could see that we have hit a plateu, in such case if we want to move further, we need to use other variants. Will it be discussed later?	Yes, when training hits a plateau or gets stuck in a local minimum, there are several go-to variants: momentum based optimizers like SGD with momentum keep some "velocity" from previous steps so the model doesn't stall on flat spots. Adaptive learning rate optimizers like Adam, RMSprop, or Adagrad adjust the step size automatically, taking bigger jumps when progress is slow and smaller ones near convergence. We can also use learning rate scheduling, lowering or cyclically varying the learning rate over time, or restarts, where you slightly bump the learning rate back up to help the model escape a shallow plateau and keep searching for a better minimum.
Is it fair to say, AI is all about Neural Network? No other model or concept behind it?	No , AI also includes transformers
after defining loss functions, how do we optimize the weights? please provide a simple example.	This is done with backpropagation, which will be discussed in details in upcoming lectures. However, as a simple example, think that the model mistakes classifying 2 times. This "loss" value of "2" should be informed to the model along with where the model did wrong, so that the model can learn to correct its mistakes.
In the context of ML/DL running multi-parameter models?	In ML and DL, models usually deal with many features at once, not just one column, so IQR gets applied per feature during preprocessing, checking each parameter separately for outliers before feeding data into the model. This matters a lot for algorithms sensitive to scale and outliers, like linear regression or KNN, since one weird extreme value in just one feature can throw off the whole prediction. For deep learning with tons of parameters and features, IQR based outlier removal is usually just one step in a bigger pipeline, often followed by normalization or standardization, since neural networks are somewhat more robust to occasional outliers but still benefit from cleaner, more consistent input ranges across all features.
Without Parameter means is it based on some rule	Not really rule-based in the classic sense, it's more that the structure comes from the data itself instead of being fixed beforehand. Take KNN, there's no equation or weight set decided in advance, it just checks the actual data points every time a new prediction is needed. Decision trees similarly grow their branching structure based on patterns found in that specific dataset. So "non-parametric" doesn't mean no logic at all, it means the model's complexity and structure aren't locked in upfront, they emerge and adapt based on however much and whatever kind of data you give it.

please explain little bittle more on ML without examples? How will that work? With some real usecases	think of how Insta is correctly predicting what you will like, youtube predicting accurately what you will watch. All this happened through ML models correctly generalizing you as a person by looking at data of several people like you
How we will identify that i reached minimal loss function ? I reached my home if i dont know its my home or not or closure to home	It is actually not possible to confirm whether the converged model is at the global minima or not. The global minima also depends on the data, so training data can have slightly different global minima compared to the testing data.
Q.) In the example of descend - the immediate descend might be shown to with light to make a decision to choose the path of descend but in the overall descend trek, there might be points where immediate ascends are found but the overall trail is descend. How is the decision making taking place when it encounters sudden ascend in the descending path?	You know you're "home" when the loss stops dropping no matter how many more steps you take, if the value flattens out over several epochs instead of decreasing, that's your sign of convergence, not a random pause. Practically, people watch the loss curve and set a threshold or patience value, like "if loss doesn't improve for 5 epochs, stop training." They also check validation loss alongside training loss, if both flatten together, you've genuinely reached a good minimum, not just memorized the training data. One catch though, sometimes you hit a local minimum that looks like home but isn't the best possible one, so techniques like learning rate scheduling or momentum help the model keep exploring a bit before settling in for good.
are non parametric and non lebeled are same	Non-parametric refers to a model's structure, it means the model doesn't assume a fixed number of parameters or a fixed functional form beforehand, like KNN or decision trees, where complexity can grow with the data itself. Opposite of this is parametric, like linear regression, which has a fixed set of parameters regardless of data size. Unlabeled refers to data, it just means the dataset doesn't have target labels or answers attached, used in unsupervised learning like clustering. Opposite of this is labeled data, used in supervised learning.
What is transformers?	A Transformer is the foundational neural network architecture powering modern Large Language Models (LLMs). Introduced in 2017, Transformers use a "self-attention" mechanism that allows the model to process entire sequences of data simultaneously rather than sequentially. This mechanism enables the model to understand the context and relationships between all words in a sentence, making it highly effective for natural language processing tasks.

Are there diff techniques to handle missing data categories like MCAR, MAR and MNAR ?	MCAR, missing completely at random, no pattern behind it, so simple fixes like mean or median imputation, or just dropping rows, work fine without much bias. MAR, missing depends on other observed columns, so smarter techniques like regression imputation, KNN imputation, or multiple imputation using MICE work well since they borrow info from related variables. MNAR, the toughest one, missing depends on the value itself that's missing, like people hiding high income. Simple imputation can badly skew results here, so this usually needs domain knowledge, selection models, or sensitivity analysis, sometimes even collecting more targeted data is the only real fix.
For the gradient descent example given, is direction assumed/already known because of the training data?	The direction is not assumed or known in advance; it is calculated mathematically during a process called backpropagation. The algorithm calculates the gradient (the derivative) of the loss function with respect to each weight in the network. The gradient points in the direction of the steepest ascent of the loss. By taking a step in the exact opposite (negative) direction of the gradient, the model ensures it is moving toward a local minimum to reduce the error.
Are the Data challenges to be taken care in both trining and testing data?	of course. outliers, null values, missing data etc are some egs
if gradient descent just blindly steps down whatever slope it's currently on, how do we know it hasn't just hit a 'local minimum' instead of the absolute lowest error? What if a completely different starting point leads to a much better optimized state? Does the algorithm actually bruteforce every possible path to be sure?	if the learning steps and learning rate are big steps and quick rates respectively this is possible so we need to play with these two parameters to avoid getting into local
why is it called imputed methods?	Imputation just means filling in missing values with estimated ones instead of leaving gaps or dropping rows entirely.
How to take care of a numerical field like AGE and CATEGORICAL part such as customer RISK as a part of Data Imputation?	for categories we use model for age we use mean or median
If we have to calculate weights, why AI gives different output even when the same prompt is entered? Do weights for the question/answer change?	Great question. Weights need to be calculated so that the model can use these weights correctly to perform the desired task. When AI chatbots give different responses to the same prompt, there is actually randomness used inside the model itself. The weights do not change, however, depending on the input, the output of the layers change when combining the input with the weights.
what is IQR? 25% 75% etc	inter quartile range

what are parametric and non parametric models	Parametric models have a fixed number of parameters decided upfront, regardless of how much data you feed in. They assume a specific functional form, like a straight line or fixed equation, and just tune the parameter values to fit the data. Examples include linear regression, logistic regression, and Naive Bayes. Non-parametric models don't lock in a fixed structure or number of parameters ahead of time, the complexity grows with the data itself. They make fewer assumptions about the underlying relationship. Examples include KNN, decision trees, and random forests.
then Raj it means i have dataset put in function get the output. In any scenario i will get less wrong answers from function then i can simply say thats my minimal loss function	Yes, you are absolutely correct. However, as the course progresses, you will gain the intuition to select the correct loss function so that you do not have to use a lot of different training sessions to arrive at the best loss function.
How can we detect outliers in categorical/non-numerical data? As per my understanding, IQR can be used for numerical data. Kindly clarify.	Categorical outliers are detected by analyzing frequencies and probabilities rather than numerical distances. Common methods include: Frequency Counts: Flagging categories that appear very rarely (for example, representing less than 1% of the dataset). K-Modes Clustering: Grouping categorical data and identifying instances that do not fit well into any formed cluster. These methods will in fact be discussed in details in the upcoming lectures.

Sir, can you provide best example to simply understand the differences between all the errors like 1. MEAN SQUARE ERROR (MSE) 2. MEAN ABSOLUTE ERROR (MAE) 3. ROOT MEAN SQUARE ERROR (RMSE) 4. MEAN ABSOLUTE PERSENTAGE ERROR (MAPE) 5. R SQUARE (R^2)	took help from LLM For this Let's use one simple example throughout: predicting house prices. Say actual prices are 100, 200, 300 (in thousands), and your model predicted 110, 190, 320. MAE (Mean Absolute Error): Just averages the absolute differences, ignoring direction. Errors here are 10, 10, 20 → average is about 13.3. Easy to interpret, tells you "on average, I'm off by this much." MSE (Mean Squared Error): Squares the errors before averaging, so bigger mistakes get punished much harder. Same errors squared are 100, 100, 400 → average is 200. Notice how that one 20-error dominates way more than in MAE, since squaring amplifies larger mistakes. RMSE (Root Mean Square Error): Just takes the square root of MSE to bring it back to the original price units. So root of 200 is about 14.1. It's like MSE but more interpretable since it's in the same scale as the actual prices, while still penalizing big errors more than MAE. MAPE (Mean Absolute Percentage Error): Converts errors into percentages relative to a
Still Not very clear on how the absolute local minimum is identified ,can this be explained in a little more detail	Basically, with neural networks you can never really prove you've hit the lowest possible point out there, there's just too much going on with all those parameters to check every corner. So instead, we look for signs that we're close enough. The main sign is the loss flattening out, if it stops going down even after training more, that usually means you've settled into a minimum. We also keep an eye on validation loss, not just training loss, because a model can fake being "good" by just memorizing the training data, validation tells us if it's actually learned something useful.
Which is a better method to find outliers? Control Charts or Boxplot?	both can be used
what is mean by IQR?	IQR is basically the "middle chunk" of your data, the range between the 25th percentile and 75th percentile, so it tells you where the bulk of typical values sit while ignoring extreme highs and lows. It's mainly used to spot outliers, anything way outside this middle range is flagged as unusually high or low compared to the rest of the data.

The topic around (irreducible) Error was interesting. As a ML devloper do we need to be bothered about (measurement) bias? If so can we really quantify it?	Yes, measurement bias is definitely something to worry about since it's systematic, not random like irreducible error, and it can quietly skew your model if unnoticed. You can quantify it by comparing your data against a trusted ground truth subset, checking for consistent patterns in residuals across specific groups or sensors, or measuring inter-annotator agreement for labeled data. Unlike irreducible error, bias is correctable once caught, which makes catching it worth the effort.
cn you telll me how IQR works , why we considers 75% data only ?	to remove outliers
does IQR always gives accurate results?	it will give generalizable results removing outliers impact
in errors vs artifaacts, what is artifacts?	Error: The measurable difference between a model's prediction (or measurement) and the true ground truth, usually caused by model inaccuracy, random noise, or statistical variance. Artifact: An artificial, unintended distortion, anomaly, or feature introduced into the data by measurement hardware, processing algorithms, or compression methods, rather than reflecting the true underlying phenomenon. For example, when a movie is compressed to a lower quality, the distortions in the video is called artifacts rather than error.
There are other outlier detection mechanism like MAD or Tukey other than IQR. WHy do we empahise on IQR in all academics and when to use the others? Can you help w/ examples?	IQR gets emphasized in academics mainly because it's simple, robust, and doesn't need assumptions about data distribution, it just uses percentiles, so it's easy to teach and works reasonably well across many datasets.
If the outliers are the truth (e.g. Ambani case explained here). Will ignoring the outliers in this case not lead to error?	outliers are hardly the truth, how many Ambanis are there. Including them to the overall, will increase the average income of India and World bank will think India is doing better than what the actual reality is
what is dataset blindspot?	dataset blind spot means there are real-world scenarios, edge cases, or subgroups that your training data just doesn't cover well, so the model never learned to handle them properly. It's not about dimensions being reduced, it's more about coverage, like training a pig image classifier mostly on daylight photos and having no examples of pigs in low light or mud, then the model performs poorly whenever it hits those unseen conditions in the real world.

Dataset blind spot means data is out of scope, that is not comprehensive with the example/training data, right?	dataset blind spot means there are real-world scenarios, edge cases, or subgroups that your training data just doesn't cover well, so the model never learned to handle them properly. It's not about dimensions being reduced, it's more about coverage, like training a pig image classifier mostly on daylight photos and having no examples of pigs in low light or mud, then the model performs poorly whenever it hits those unseen conditions in the real world.
when to use low vs high dimension for reducing blindspots	Low dimension, like after PCA, helps cut noise and redundant features, good for speed and generalization, but going too low risks losing subtle signals and creating blind spots. High dimension keeps more detail, better for catching rare or subtle patterns like fraud or medical anomalies, but raises overfitting risk and needs more data to support it.
in this highdimension view was obstructed, else it shud be better right ?	Correct.
what is mean by PCA? can you please use full forms rather than abrevations	PCA is short for "principle component analysis". It is used to select combinations of features which give the best performance when a subset combination of the full set of features are used.
Should we always reduce dimensions? Or in some cases higher dimensions make sense	no not necessary : Repeating this ans : Low dimension, like after PCA, helps cut noise and redundant features, good for speed and generalization, but going too low risks losing subtle signals and creating blind spots. High dimension keeps more detail, better for catching rare or subtle patterns like fraud or medical anomalies, but raises overfitting risk and needs more data to support it.
is it always advised to break down to low dimension? But that will be very difficult. ?	While reducing the dimension is used for efficiency, it is indeed NOT always advised to reduce dimensions, as it can also reduce the accuracy of the model.
How do AI companies ensure that their new model will be better than the old model? Why it cannot be worse? Someone told me that the new model is built from scratch and not from the old model.	Commpanies run extensive benchmark evaluations comparing both models side by side on the same test sets covering things like reasoning coding safety and factual accuracy before any public release. It technically can be worse on some specific tasks even if better overall that's why companies track many benchmarks not just one score and often keep the old model available too in case some workflows still perform better on it. So "better" isn't guaranteed universally it's usually "better on average across the benchmarks that matter most for the intended use

What is degree in the function?	degrees of freedom you mean? here is what it means : if you have 5 numbers and you know their average, only 4 of them can be anything you want, the 5th is automatically fixed to make that average true. So degrees of freedom here is 4, not 5, since one value lost its "freedom" to the constraint.
if we reduce dimensions, shouldn't it also remove some information? so how can we find out what dimensions are to be removed with minimal information loss?	Great question. Depending on the data, reducing the dimensions can in fact reduce important information as well. Feature selection and principle component analysis can be used to reduce dimensions while ensuring high accuracy. These concepts will be covered later in the course.
parameter as in degree of x , but then the parameters are interdependent	yep you are right as the degrees increase there could be overfitting as if one changes others change too
Does PCA is preferred - as it reduced the variability to some extent - how much cardinality can we reduce keeping the optimum variance	PCA reduces dimensionality (the number of features), not cardinality (the number of unique values within a specific feature). To maintain optimum information, a standard practice is to retain enough principal components to explain 95% to 99% of the cumulative variance. The exact number of dimensions you can safely eliminate depends entirely on how highly correlated your original features are.
if we reduced dimensions isn't there a possibility to loose information as well? so how do we identify what dimensions to reduce to make the information loss as low as possible?	Yes, some information loss is inevitable, but PCA picks directions of maximum variance first, so it drops the dimensions that carry the least useful signal, not random ones. Like in a photo, PCA would keep the components capturing overall shape and lighting, and drop tiny pixel-level noise that barely affects the picture. but you can control the information loss
so if a definite algo exists for a problem, does it called rule-based? and ai isnt needed to solve it?	not really , not all algorithms are rule based, they learn by optimizing loss functions. AI is not always needed, but doesn't mean the regression classification problems are rule based
MSE?	mean squared errpor
failing in inference meaning taking long time because of more dimensions?	Failing in inference due to high dimensions refers to the computational bottleneck known as the "curse of dimensionality." When a model has thousands of features, generating predictions (inference) takes significantly more processing power, memory, and time. This computational burden can cause real-time systems to lag, time out, or crash entirely during production.
how to identify the minimum parameters in a 100 parameter set? Only trail and error and with statistics? or any easier method?	there are algorithms that help us identify top variables that help explain the variability in the data. Like Lasso regression
Why Testing is bad since it has same kind of data set like training data ?	testing dataset should be outside the time frame, maybe more recent so that we know how well model is performing in recent data. It is not bad, it is necessary

what are the disadvantages of overfitting? and shouldnt it be ideal given that it gives comparatively best output	While overfitting means that the model fits even perfectly to the training data, this also means that the model will perform very bad on the real-world data that it sees after the training has already completed. Therefore, overfitting makes the model unfit for use practically.
I am following the professor without having any ml background, do I need to practice in python also or concept in this session is enough to follow the upcoming trainings?	As long as you study the topics from this lecture, you should be able to follow the upcoming lectures. However, you should in fact practice the corresponding python experiments, otherwise it will be very difficult to follow the demo sessions.
what is the difference between validation data and test data?	Validation data is used during training to tune the model like a new set of data to adjust hyperparameters or deciding when to stop training, you check performance on it repeatedly while still building the model. Test data is used only after the model is fully built and finalized on the truly unseen data that tells you how the model will actually perform in the real world.you don't touch it or tune anything based on it.
When Training Data is fitting good and Validation/Test Data is fitting bad, should we try and optimise the same model or switch to a different model?	In this case, the model should be tweaked, perhaps changing the number of layers, number of neurons per layer, learning rate, loss function or a combination of these parameters.
Does this mean, overfitting can happen due to multiple feature values as input for a supervised model? Should we remove the less important features at the time of training to avoid overfitting?	overfitting happens as the multiple powers of same inputs are used. Yes not using a lot of them helps reduce the overfit
why K fold term used?	to cross validate across different datasets to see if the model continues to be robust
except for CV , is there any other method , to reduce underfitting and overfittig , and how ?	using linear models to reduce overfitting, you can use regularization, drop out , feature engineering as other ways to optimize the fit
What approach we will take to divide data in k-cross validation method?	randomly exclude 10% of data K times with or without replacement, training the model on included data and test it on excluded data, keep repeating K times and evaluate the resutls
And to follow up will sorting the data on a particular parameter and split for training and testing bring the difference in model fitting?	Yes, and it will make the testing performance worse. For example think of flower images used to detect the species of flower. If petal size is used for sorting, then the training subset can have the smaller petal sizes while the testing subset can have the larger petal sizes. So the model gets trained only on smaller petal sizes, performing very bad on testing data afterwards.
Is the confusion matrix evaluation for classification problem?	yes to evaluate TP, FP, TN, FN and the ratios around them like recall, precision etc

Sir, I have 2 questions Q1. Can you explain these below errors with good examples, something like example given for gradient descent. 1. MEAN SQUARE ERROR (MSE) 2. MEAN ABSOLUTE ERROR (MAE) 3. ROOT MEAN SQUARE ERROR (RMSE) 4. MEAN ABSOLUTE PERSENTAGE ERROR (MAPE) 5. R SQUARE (R^2) 2Q. How should we understand, when should we use which function as one function is not suitable for all.	Given actual values Y = [10, 20] and predictions [12, 15] (errors: 2, -5): -> MSE: Averages squared errors. Result: 14.5. Use to heavily penalize large errors. -> MAE: Averages absolute errors. Result: 3.5. Use to handle outliers without skewing weights. -> RMSE: Square root of MSE. Result: 3.8. Use to penalize outliers while keeping original units. -> MAPE: Absolute error as a percentage. Result: 22.5%. Use for intuitive, scale-independent reporting. -> R-squared: Proportion of variance explained. 1.0 is perfect; 0.0 predicts the mean. Use to evaluate overall model fit.
I am very new to the concept. How to get the model designed/ train the model.	This will be covered in a lot more detail in the upcoming lectures. There are dedicated lectures for designing and training different models.
Can we ask profressor to take some real-world example of why overfitting is infact not a good idea from practical usage purpose after training? Many thanks	Sure, after the coding session, you can raise your hand and directly ask this question to professor. But one example can be that say there are a small number of erroneous labels in the training data (labelling cycle as a car). When overfitting happens, the model actually memorizes to detect the cycle as a car, ultimately failing in cases were a similar cycles are seen in the real-world.
Do all the ML algorithms work on basis of probability? True Positive, False Negative sounds like a probability concept. As per my knowledge, AI is something that uses probability rather than pattern to make token predictions, so is it the same case in terms of Machine Learning as well?	you cannot generalize like that , sure many algorithms uses prob like logistic reg, NN, naive bayes, even LLMS. but think of K means , SVM, decision trees etc, they use linear algebra
how can FN be bad for cancer. its not cancer and model said no for same ?	FN means in truth they had cancer but model said no, that is dangerous no? Those patients won't get treated
I have different opinion. Actually the classification it's quite dependent on the objective. For example, if the patient has a brain tumor in that case FP can lead to invasive procedure. However, in the case of skin cancer FP does not matter as much as the brain tumor case.	still unnecessarily worrying a patient by telling they have cancer when they don't is not good no?
shouldn't it be tp+fn / total , as tp is considered true and fp is considered true as well?	recall is TP/(TP+FN) as we need to know off those with cancer how much the model actually predicted correctly. What you are giving is actual positive rate.
what r rule based algo?	decision trees which uses gini coef values as a rule to determine purity and split the data accordingly

Which on is better metrics to decide on the robustness of the model? Accuracy/Precision/Recall/F1-Score??	Depends on your problem statement and dataset.
Can you please explain about the last question asked where Yes and No were 50-50%	Yes, the answer is yes, in case of unsupervised learning model learns without labels
when to use supervised learning and unsupervised learning?	when you know the Y values like it rained or not, they got cancer or not , the temperature is so much then use supervised, when you want to categorize from the invoice data who are bargain hunters vs who are premium buyers that is unsupervised
so for supervised learning we dont need labelled data ??	For supervised learning we need labelled dataset, for unsupervised we dont
so we cannot say reducing FN is the right objective for all kind of cancers (kind of generalisation)? it depends on cost of doing mistakes which would vary based on the use case as there is likely be trade off between precision and recall.	yes cost from all perspectives should be considered. in healthcare example it is expensive to get the doctors to be convinced. So the pharma companies send doctors on conferences which are expensive. So they choose only doctors with high likelihood of prescribing. If they miss even one of them, they have lost a lot of potential sales
where do we uses reinforcement learning	Whenever the losses are too complex and very poorly defined, reinforcement learning works best. For example, working more for an appreciating boss rewards well, but for a boss that starts giving unbearable burden, does not reward well. And life is filled with problems like this where the ultimate success is very poorly defined. Reinforcement learning is used in this case. In fact learning for us humans is basically reinforcement learning.
Which learning algorithm do we use in robotics?	reinforcement learning is used for sure
so in reinforcement learning, i am looking at my loss function and optimizing weights to minimize the loss function in each iteration?	In reinforcement learning, there's no "correct answer" given upfront. Instead the robot takes actions in an environment and gets a reward (or penalty) as feedback. The goal becomes maximizing cumulative reward over time not just minimizing a straightforward error against a known label. what's being optimized is expected future reward, not a direct error signal like in supervised loss functions.
From the ethical perspective, I would worry more about patients than pharma companies. but again it's a philosophical debate :)	yes I gave that example to help determine another way to calculate cost
What can be done with the data output from unsupervised learning models? For example, in scenarios where children identify and group different flower types.	it will help e.g Netflix understand what groups of folks to target for a specific movie recommendation. You as a person maybe similar to a person from US which only data shows , so whatever he watched you and others like him will be shown that movie as a recommended movie to be watched

but i can use k cross validation in reinforcement learning. No?	Reinforcement learning data is generated sequentially by an agent interacting with an environment, which violates the independent and identically distributed (i.i.d.) assumption required for standard cross-validation. Evaluating an RL policy requires running it in an active environment to measure cumulative rewards, rather than testing static predictions against a holdout data set. -> Online RL: Incompatible with k-fold cross-validation because the data distribution constantly changes as the policy updates. -> Offline (Batch) RL: k-fold cross-validation can be applied here to evaluate value function approximations because the dataset is fixed and pre-collected. -> Standard Practice: RL agents are evaluated by averaging cumulative rewards over multiple test episodes across different random environment seeds. Your premise that k-fold cross-validation can be used in reinforcement learning is incorrect for standard online training, but it is correct for offline batch settings.
So, reinforcement learning should take more time and energy to complete a task in comparison with supervised and non-supervised learning ?	yes but we have Monte carlo simulation and other such probabilities simulation techniques to help us design it better without exposing it to real world data
In RL, especially for intermediary steps is there a reinforcement feedback?	yes it is called as reward shaping
how supervised learning different from reinforcement learning. supervised learning also adjusts rge parameter to meet the target automatically.	for reinforcement we train based on outputs in the intermediary steps itself , do not have to wait for the entire target prediction
how supervised learning different from reinforcement learning. supervised learning also adjusts rge parameter to meet the target automatically.	Supervised learning know the labels before start. Reinfrocement learning have the answer after a sequence of step
What is index_col?	Just serial numbers
how long will the class continue?	1>30
How reinfocement learning get feedback	Do trial and error initially. Asnit learns, the sequence of steps will be sensible
How system knows what is sensible , might the objective of robot is to fall erather walk	Yes. Also, less energy expended.
How to remove duplicates if exist in data	Some simple correlation will do. Also there are functions to get duplicates
Sitting idle will be least enery and shortest path - How robot decioded that my objective is to walk	It is AND rule. Reach to a destination AND expend much energy
what to do in case of duplicate rows found after we remove unique column ID? how you are identifying duplicate rows in this case study?	Need to remove duplicate rows before all preprocessing (removing some cols)

Soyry to bug you but still reaching to destination while walking is least enery than running , how robot decided to lurn to run or jump	The objective is you define: E.g., reach a destination + Lower energy expended — May be robot walks as running expend more energy. 2. Reach destination + less time: RObota may run
ok so how objective is defines in mathmatical terms, I can relate to supervised learning where labelled data is objective	This will be obtained from actuators/sensors
I am under the impression that we do not look for outliers in a supervised learning. is that true?	No, Outliers has an important problem as discussed
Actuators and sensors will provide the state , what stae is desired , is this also defined in actuator iand senser ? This is we are talking in terms of Robot , but how the objective defined for let say image or language	Robots havemultimodal sensors. What we require need not use images compulsoriy. We can get info from battery transferred to a computer
what do you mean by imputing an outlier?	Identifying outlier based on an attribute. In a given data point, obly one attribute will be outlier. TO save the datpoint, we need to fill that outlier sensibly. THat is imputation
Objective - I want to build model that becomes painter like picaso , how will I define objectice ?, Will it supervised or unsupervised or reinforment learning	It can be supervised. Like in sequence model or NLP. Why because at the end of sequence of action, what objectivity tell the success/failure
Is it required to standarize/normalize/carry out data transformation across both features as well as target? or only feature is enough and why?	both please
how does random_state parameter work?	In ANN, we have parameters. Initializiation is done at the start for optimization. If make random state, then it will use the same initialization again
how does random_state parameter work?	The random_state parameter dictates the value of random seed to use. For example, if no random_state is given, then each time the code is run, the output is different. The random state makes the output reproducible. Different random states make the random functions use different random parameters, but still in a predictable way.
Won't filling the outliers with Median value make the dataset biased?	Yes, but if the outleirs are less, median is better than zero
median is not dependent on outliers, so can you please expalin why are we negating the outlier mask to find the median?	Outlier tells that datapoint is not good, and adverse effect on model. For the outlier attribute, we need to remove the outlier value. A good case we can do is median or mean, which by statisti cs will not be rare.
I understand imputation for missing values, but why is it necessary for outliers, can't they be simply removed?	We are losing datapoints which are costly
How will I define label for abstract painting that never created - I am trying to build my nderstanding - Apologies for naive questions	Like natural language generation, the previous strokes determine the future. Need to use generative AI concept.

How will I define label for abstract painting that never created - I am trying to build my nderstanding - Apologies for naive questions	This is actually a great quesiton. It is generally acheived with adversarial loss. Basically, instead of one model, two different models are used. One model tries to produce abstract painting, while the other model tries to detect which is the real abstract painting and which is the generated one. By trying to fool the detecting model, the generating model becomes good in changing the look to an abstract painting. These types of networks are called generative adversarial networks.
StandardScaler is directly mapped to the correlation function - scale and then find the correaltion - is it always linear correlation - or positive or inverse proportion?	it does pearson correlation ranging from -1 to 0
StandardScaler is directly mapped to the correlation function - scale and then find the correaltion - is it always linear correlation - or positive or inverse proportion?	it does pearson correlation ranging from -1 to +!
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what does corrleation is not causation mean ? Are there causal/correlated feature here to be considered for modelling?	e.g during summers both ice cream sales and tender coconut sales increase. ice cream and tender coconut are correlated here , they do not cause each others sale rise. the underlying factor is summer
what does corrleation is not causation mean ? Are there causal/correlated feature here to be considered for modelling?	Crimerate in IIT is low because secutities are high alert. Vws Securities are high alert because Crimerate in IIT is low. First is Causation = Correlation. Second Causation not equal to correelation
or the correlation hides the higher degree interrelation?	Good question.But a correlation is a 2 order statistics, which is computationally tractable.
how to decide which columns to drop?? Do we need to have domain knowledge for that or do we any other way to find it?	Domai knowlege and correlation metricbased
Is filling outlier with median a common step to take while training and testing ML model? Or are there exceptions?	it is the first step yes
Why we use the same median/imputed function for test data?	Indistribution. Otherwise test data will be OOD
What is the need of scaling the attribute to same scale while predicting? what happens if we keep the attributes as is?	the coefficients will be very high in proportional to the magnitude of the values in X. So it will lead to wrong assumption on importance and coefficients won't be apples to apples comparison
As this is a classification problem, Why are we not using Logistic Regression and going with Linear Regression?	We can use both. The difference you will see in next lecture
Not sure about others but for me this going over head. what can I do to be more familiar with all of these coding concepts? Do I need be more expert on python libraries as well?	Pl practice with our amterials
Isn't F1 a better evaluation metric for model ? Why do we need to focus on Precsion or recall to evaluate the model performance?	F1 is for optimising both FP and FN. Depending on the business (as discussed in other ans already) we would need to optimize for one of them only

How does it impact, if we dont use the same mean/median of training data for test data? how test data will be OOD if we dont use this logic?	-> Computing a separate mean or median for the test set applies a different mathematical transformation than what the model learned, which introduces data leakage and inconsistency. -> A machine learning model establishes its weights and decision boundaries based entirely on the specific statistical distribution (the baseline and variance) of the training data. -> Imputing or scaling the test set using its own internal statistics physically shifts its feature space away from that learned training baseline. -> This shift artificially turns the test data into Out-of-Distribution (OOD) data, as the model is now forced to evaluate inputs centered around a completely unfamiliar mathematical baseline, which breaks the learned patterns and ruins predictive accuracy.
Hi, In section 2.1 of the code base shared, diagnosis was dropped as a feature from matrix X. How is stratify working after that step to derive _train and _test matrices?	Because we are passing y which contains the labels
Sorry my mic isn't working for some reason. I'll ask the question in the here. 1) Can the loss function be a user defined equation ? 2) Related to Unsupervised learning : Is there any other use cases of unsupervised learning apart from clustering.	1. Yes absolutely, provided it can support learning through backpropagation.
Sorry my mic isn't working for some reason. I'll ask the question in the here. 1) Can the loss function be a user defined equation ? 2) Related to Unsupervised learning : Is there any other use cases of unsupervised learning apart from clustering.	1. Yes, provided the loss function supports backpropagation. 2. Any task where the output is not clearly defined. For example, changing a real photo to look like a painting might not have all input-output pairs, but some real and some paint images.
how to summarize these things , so that we can prepare for assignments	See its like, preprocessing and evaluaation metrices almost remains same with some changes here and there depending on the data, only the model changes, so after 2-3 sessions you will develop the intuition.
Labelling of data is required?	For supervised learning, yes. For unsupervised learning, no.
When should we do the scaling of features?	When different features have too different range of values. For example, [-1, 1] vs [100, 900]
I am working on a classification problem. The feature list is a mix of categorical and numerical features. However, the categorical features are dominating xgboost and the Recall has fallen down. Will removing the categorical features would help over here? The prob is related to banking domain	Do not remove the categorical features, as they likely contain highly valuable domain information. Tree-based models like XGBoost often struggle with high-cardinality categorical variables when they are one-hot encoded, as this creates highly sparse data and fragmented trees that fail to generalize. Instead, use algorithms natively designed for categorical data, such as CatBoost or LightGBM.